Yong Guk Kang

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Summary

Research Professor at Seoul National University's Imaging Intelligence Laboratory, working on computational imaging, interferometric imaging, biomedical imaging, lensless imaging, quantitative phase microscopy, optical coherence microscopy, nonlinear microscopy, and biomedical image analysis. My work connects optical system design, wave-optics simulation, prototyping, and quantitative analysis for interdisciplinary imaging problems.

Research Interests

Computational optics, Optical imaging systems, Lensless imaging, Quantitative phase microscopy, Biomedical photonics

Education

Ph.D. in Biomedical Engineering, Korea University (2014-03-2020-08)
Thesis: Development of a multiphoton and optical coherence microscope for visualization and analysis of cell-extracellular matrix interactions

- Implemented live-cell multiphoton and optical coherence microscopy for cell and tissue imaging.
- Analyzed 3D matrix deformation using digital volume correlation.

B.S. in Biomedical Engineering, Korea University (2008-03-2014-02)

Positions

Research Professor, Seoul National University, School of Mechanical Engineering (2024-04-present)

- Designed and simulated transparent interactive display systems with integrated camera functionality.
- Developed lensless imaging systems and wave-optics simulation workflows.
- Applied machine learning based inverse design methods for optical elements.
- Fabricated optical elements using DMD light-engine based grayscale lithography workflows.

Postdoctoral Researcher, Korea University, School of Biomedical Engineering (2022-09-2024-03)

- Implemented successive accumulation of interferogram based 3D reflection phase microscopy.

- Developed sparse axial scanning and virtual refocusing strategies for efficient 3D imaging.
- Designed relay optics for laser scanning systems.
- Built ZEMAX-based 4-f relay optics from off-the-shelf components.

Research Lecturer, Korea University, School of Medicine (2020-10-2022-08)

- Developed optical coherence microscopy methods for quantifying cell-ECM interactions.
- Applied temporal dynamic contrast imaging to analyze intracellular motility.
- Simulated Bessel/Gaussian beam switchable OCM to extend depth of focus.

Selected Publications

Yong Guk Kang and Kwanjun Park and Min Gu Hyeon and Taeseok Daniel Yang and Youngwoon Choi. **Three-dimensional imaging in reflection phase microscopy with minimal axial scanning**. *Optics Express*, 2023.

Yong Guk Kang and R. J. E. Canoy and Yujin Jang and A. R. M. P. Santos and Inseong Son and Beop-Min Kim and Yongdoo Park. **Optical coherence microscopy with a split-spectrum image reconstruction method for temporal-dynamics contrast-based imaging of intracellular motility**. *Biomedical Optics Express*, 2023.

Kwanjun Park and Taeseok Daniel Yang and Yong Guk Kang and Taeyoon Kong and Youngwoon Choi. **Wide-area topography using reflection phase microscopy for surface inspection**. *Optics Express*, 2025.

Yong Guk Kang and Donggeon Bae and Nakkyu Baek and Hyojun Ahn and Taeyoung Kim and Kyung Chul Lee and Seung Ah Lee. **Non-Paraxial Layered Diffraction Simulator for Inverse Design of Lensless Imaging Optics**. *Optica Imaging Congress 2025*, 2025.

Hyesuk Chae and Kyung Chul Lee and Yong Guk Kang and Jongho Kim and Kyungwon Lee and Suki Kang and Geunbae Bang and Nam Hoon Cho and Seung Ah Lee. **High-speed virtual re-staining via near-infrared quantitative phase imaging**. *Proceedings of SPIE Advanced Biophotonics Conference*, 2026.

Kyung Chul Lee and Hyesuk Chae and Jongho Kim and Lucas Kreiss and Hyeongyu Kim and Yong Guk Kang and Kevin C. Zhou and Amey Chaware and Kanghyun Kim and Shiqi Xu and Suki Kang and Geunbae Bang and Nam Hoon Cho and Dosik Hwang and Roarke Horstmeyer and Seung Ah Lee. **Semi-supervised virtual staining using learned-illumination Fourier ptychography for high-speed label-free histopathology**. *Journal of Physics: Photonics*, 2025.

Yongjun Jang and Myeongjin Kang and Yong Guk Kang and Dongtak Lee and Hyo Gi Jung and Dae Sung Yoon and Jongseong Kim and Yongdoo Park. **Cardiac fibroblast-mediated ECM remodeling regulates maturation in an in vitro 3D engineered cardiac tissue**. *Journal of Tissue Engineering*, 2025.

Yong Guk Kang and Hwanseok Jang and Yongdoo Park and Beop-Min Kim. **Development of a 3-D Physical Dynamics Monitoring System Using OCM with DVC for Quantification of Sprouting Endothelial Cells Interacting with a Collagen Matrix**. *Materials*, 2020.

Yong Guk Kang and Hwanseok Jang and Taeseok Daniel Yang and Jacob Notbohm and Youngwoon Choi and Yongdoo Park and Beop-Min Kim. **Quantification of focal adhesion dynamics of cell movement based on cell-induced collagen matrix deformation using second-harmonic generation microscopy**. *Journal of Biomedical Optics*, 2018.

Hyun-Joong Kim and Se-Hwan Um and Yong Guk Kang and Myeongjin Shin and Hyeon Jeon and Beop-Min Kim and Dongtak Lee and Kisoo Yoon. **Laser-tissue interaction simulation**

considering skin-specific data to predict photothermal damage lesions during laser irradiation. *Journal of Computational Design and Engineering*, 2023.

Min Gu Hyeon and Taeseok Daniel Yang and Jong Su Park and Kwanjun Park and Yong Guk Kang and Beop-Min Kim and Youngwoon Choi. **Reflection Phase Microscopy by Successive Accumulation of Interferograms**. *ACS Photonics*, 2019.

Taeseok Daniel Yang and Kwanjun Park and Yong Guk Kang and Kyoung J. Lee and Beop-Min Kim and Youngwoon Choi. **Single-shot digital holographic microscopy for quantifying a spatially-resolved Jones matrix of biological specimens**. *Optics Express*, 2016.

Projects

Lensless and HDR computational imaging. Development of lensless imaging systems for high dynamic range applications in extreme lighting conditions, including non-paraxial wave simulation and inverse design of optical elements.

Reflection phase microscopy. 3D reflective phase microscopy using successive accumulation of interferograms, sparse axial scanning, virtual refocusing, and field synthesis.

Optical coherence microscopy for cell-ECM dynamics. Optical coherence microscopy methods for live-cell and tissue imaging with temporal dynamic contrast.

Live-cell nonlinear microscopy for matrix deformation analysis. Second-harmonic generation and two-photon microscopy workflows for visualizing collagen matrix deformation and cell-ECM mechanical interactions.

Optical prototyping and fabrication. Rapid prototyping for optical and biomedical experiments, including FDM 3D printing, PDMS microfluidic chip fabrication, and DMD grayscale lithography for planar optical elements.

Funding, Fellowships, and Patents

Funding: Creative and Challenging Research Base Support, National Research Foundation of Korea (2021R1I1A1A01048370)

Fellowship: Global Ph.D. Fellowship, National Research Foundation of Korea (2015H1A2A1030048)

Patent: Head mount system for providing surgery support image, United States Patent (US 11,356,651)

Patent: Reflection phase microscopy system for 3D image acquisition, Korea Patent (10-2024-0081557)

Patent: Phase mask fabrication method based on grayscale lithography, Korea Patent (10-2024-0161144)